02.05.2025

The Research Council of Finland

# Letter of Collaboration

To whom it may concern,

I am writing to confirm that I have agreed on our research collaboration with the project **"Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)"** lead by Prof. Maarit Korpi-Lagg from Aalto University, submitted to the *"2025 Special funding for international collaboration in high-performance computing"*.

Given the goal to develop AI-assisted simulation tools to model magnetized plasmas, I foresee many challenges in optimizing computing resources, in terms of computational costs and performance, and data resources, in terms of managing and analyzing huge output data volumes. I support the research focus on reducing computational costs, development time, and storage needs. These topics fit very well with my expertise and my team as, over the last many years, I have researched performance observability, cloud and HPC, big data analytics (workflows and streaming analysis) and recently opportunistic data operations optimization in cloud-HPC continuum. Prof. Korpi-Lagg and I were co-PI of the NEOSC subproject in the Inno4scale EuroHPC project, in which my team has developed the opportunistic data analysis platform (ODOP). In this project collaboration, I will support Prof. Korpi-Lagg team to utilize and improve ODOP for MomEnt. Especially, we will not only collaborate on the features of utilizing the idle CPU resources for orchestrating the GPU computations and ML tools for other vital data analysis processes to collect standard data products from the main simulation on the fly, but will also further develop ODOP for versatile and complex HPC workflows in MomEnt.

My team and I will also contribute to techniques and methods for the software prototype and integration of HPC data analysis and resource optimization and workflows. I will have a common supervision of the PhD student and postdoc working on ODOP and HPC workflows. Since Prof. Korpi-Lagg and I are in the same department in Aalto and we have conducted many joint works, I will work closely with Prof. Korpi-Lagg team. I will also provide opportunities for Prof Korpi-Lagg's team members to learn our research activities in our core areas.

Sincerely,

Hong-Linh Truong, Associate Professor

Aalto University
Dept. of Computer Science
Hong-Linh Truong

Postal address
P.O. Box 15400
FI-00076 AALTO

Visiting address
Konemiehentie 2
Espoo, Finland

Tel +358 50 407 5422

https://users.aalto.fi/~truongh4/
linh.truong@aalto.fi

4.5.2025

## Letter of Collaboration

This is a letter of collaboration for the Academy Professor project proposal “Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)” of Assoc. Prof. Maarit Korpi-Lagg from Aalto University. I am a Professor of Parallel and Distributed Data Science, specializing in parallel computing methods for big data processing and parallel GPU algorithms. I am very interested in collaboration on GPU-accelerated stencil algorithms, as efficient concurrency control of GPU computing is one of my key research areas. We have already done a joint publication “SOMA: Observability, monitoring, and in situ analytics for Exascale applications” which focused on the analysis on simulation data from massive GPU based HPC simulation using in-memory data streaming between the main simulation and a data analysis application opportunistically using computing CPU resources unused by the GPU simulation. I am very interested in continuing our collaboration also in the context of this new MomEnt proposal, where our main focus will be integration and optimization of the big data analysis workflows to LUMI supercomputer.

Prof. Keijo Heljanko

Deparment of Computer Science

University of Helsinki

Finland

Email: keijo.heljanko@helsinki.fi

Phone: +358503084871

THE UNIVERSITY OF TEXAS AT AUSTIN

Arya Farahi, *Assistant Professor*
*Department of Statistics and Data Sciences*
*E-mail: arya.farahi@austin.utexas.edu*
*URL: https://afarahi.github.io/*

**To**: Research Council of Finland (RCF)
**Project Title**: Modelling Turbulent and Magnetised Plasmas at Exascale with AI-assisted Simulation Tools (MomEnt)
**Principal Investigator (PI)**: Prof. Maarit Korpi-Lagg
**Call**: Special Funding for High-Performance Computing (HPC)

**Collaboration Letter**

Dear Evaluation Panel,

I am pleased to confirm my collaboration on the *Modelling Turbulent and Magnetised Plasmas at Exascale with AI-assisted Simulation Tools (MomEnt)* project, led by Prof. Maarit Korpi-Lagg. This proposal, submitted to the Research Council of Finland's Special Call for High-Performance Computing, represents a bold and timely effort to push the boundaries of computational astrophysics.

The project addresses one of the most complex challenges in solar and plasma physics: understanding the magnetic dynamo processes of the Sun through the utilization of large-scale, GPU-accelerated simulations. By employing exascale computing, we aim to develop a new generation of AI-powered Plasma simulation that brings new insights into solar magnetism, an understanding essential for predicting Space Weather hazards affecting the Earth. Numerous critical infrastructures, such as satellites, electrical power grids, and communication systems (e.g., navigation via the Global Positioning System; radio communications), remain susceptible to these hazards. Furthermore, in addition to the creation and behavior of plasma observed in natural settings, artificially generated plasmas in laboratories possess a diverse array of technological applications. In particular, Plasma research is crucial in fusion generators where magnetic effects need to be understood to enable the successful production of energy.

Beyond raw computational performance, this project addresses critical challenges in managing petabyte-scale data streams, optimizing AI training workflows, and maintaining a simulation-analysis pipeline that is both high-throughput and interpretable. My group is committed to advancing these areas, aiming to convert computational complexity into new scientific capabilities that align with international priorities in AI, high-performance computing, and solar physics.

I am an Assistant Professor at the University of Texas at Austin and serve as co-PI and AI Lead at the NSF-Simons AI Institute for Cosmic Origins (CosmicAI), established in October 2024. CosmicAI develops advanced AI methodologies to investigate fundamental questions about the origins of the universe. It brings together a consortium of leading institutions, including UT Austin, the University of Utah, the University of Virginia, the National Radio Astronomy Observatory (NRAO), NSF's NOIRLab, Stanford University, and UCLA. The institute's research centers on four core AI themes: trustworthiness, efficiency, interpretability, and robustness. Additionally, it fosters collaboration through open-source tool development, shared data resources, and AI education initiatives aimed at broadening AI accessibility in science communities.

My contributions to this project focus mainly on scalable machine learning, with emphasis on physics-informed neural networks and real-time simulation analysis. In collaboration with Prof. Korpi-Laggs group, we aim to develop AI-driven techniques for efficient interpretation of large-scale solar DNS datasets. A primary goal is the development of robust subgrid-scale models that maintain physical fidelity while reducing computational demands, which is critical for sustainable high-performance computing. Our collaboration will include biweekly virtual meetings, joint development of machine learningaugmented simulation tools, and co-authored publications. Subject to feasibility and funding, we also plan to coordinate research visits and workshops to deepen collaboration. Please note that my participation in this effort does not entail any request for financial support from the RCF.

I support this proposal and look forward to working with Prof. Korpi-Lagg and the Aalto University HPC Lab. Jointly, we have the potential to transform simulations that can potentially revolutionize our understanding of the Sun and push the boundaries of computational astrophysics.

Sincerely,

Prof. Arya Farahi
Assistant Professor
Co-PI, CosmicAI Institute
AI Lead, Explainable Universe

April 30, 2025

**Letter of Support**

To whom it may concern:

I am writing on behalf of NVIDIA Corporation to endorse the application "**Modelling turbulent and magnetised plasmas at Exascale with AI-assisted simulation tools (MomEnt)"** to the **Research Council of Finland (RCF) – 2025 Special funding for international collaboration in High-Performance Computing**, led by professor Maarit Korpi-Lagg.

NVIDIA is greatly invested in developing state-of-the-art hardware and software solutions in the fields of accelerated computing and Artificial Intelligence (AI). We see physical sciences becoming a domain that can significantly benefit from AI, as demonstrated in the recent advances in AI-based weather forecasting models which can deliver accurate predictions orders of magnitude faster than their traditional counterparts. NVIDIA is also developing various tools to support the convergence of numerical simulations and AI, including NVIDIA PhysicsNeMo, an open-source deep learning framework for Physics-AI models, NVIDIA Warp, a framework for building accelerated and differentiable simulation code, and TorchFort, a C/Fortran library that allows strong coupling of AI frameworks and numerical simulation HPC codes.

The foundational research goal of MomEnt is to advance AI-accelerated solvers for plasma simulations, as classical HPC methods with increasing mesh sizes are becoming intractable due to explosion of data. The group aims to achieve this by undertaking large scale high-fidelity simulations from which the AI models can learn the interaction between complex physics at the smallest scales and physics at larger scales. When trained, they can be run in conjunction with new numerical simulations to improve their accuracy and stability at more tractable resolutions.

NVIDIA has had the opportunity to support Prof. Maarit Korpi-Lagg's research, under the umbrella of the NVIDIA AI Technology Center (NVAITC) Finland and can attest of the quality of the work carried out at the group. We look forward to collaborating with the group on the aforementioned proposal. Please note that while NVIDIA has an intent to help, any formal collaboration is subject to mutual written agreement signed by the parties and NVIDIA's plans are subject to change.

Best regards,

Marc Hamilton
Vice President, Solutions Architecture and Engineering
NVIDIA Corporation